

CSE 402

University of Asia Pacific

Course Name : Operating System Lab

Lab Report 6

Topic : **FIFO Page Replacement Algo**

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Problem Statement :

Consider the following page reference string: **[1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5]**.

Implement the **FIFO (First-In, First-Out)** page replacement algorithm with a Frame Size of 4. Calculate the total Page Faults, Page Hits, and Page Hit Ratio. Analyze the results to discuss Belady's Anomaly, a phenomenon where increasing the number of page frames paradoxically increases the number of page faults.

Source Code :

```
def fifo_page_replacement(reference_string, frame_size):
 frames = []
 page_faults = 0
 page_hits = 0
 pointer = 0

 print(f"FIFO Page Replacement (Frame Size = {frame_size})")
 print(f"Reference String: {reference_string}\n")

 print("Step\tPage\tFrames\t\tStatus")
 print("-" * 45)

 for i, page in enumerate(reference_string):
 if page in frames:
 page_hits += 1
 status = "Hit"
 else:
 page_faults += 1
 status = "Fault"
 if len(frames) < frame_size:
 frames.append(page)
 else:
 frames[pointer] = page
 pointer = (pointer + 1) % frame_size

 print(f"{i+1}\t{page}\t{frames}\t\t{status}")

 print("-" * 45)
 print(f"Total Page Faults: {page_faults}")
 print(f"Total Page Hits: {page_hits}")
 print(f"Page Hit Ratio: {round(page_hits / len(reference_string), 2)}")

reference_string = [1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5]
frame_size = 4

fifo_page_replacement(reference_string, frame_size)
```

Output Screenshot :

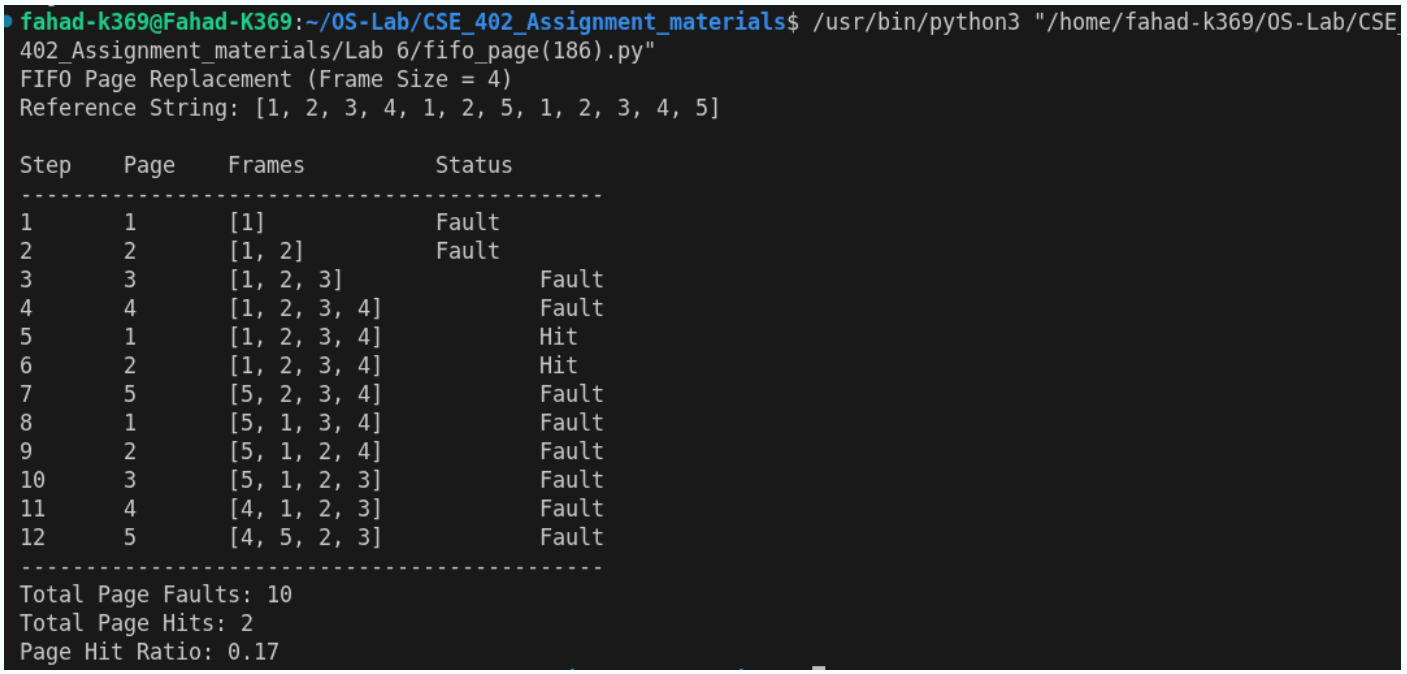


Fig : FIFO Page replacement

Discussion :

The simulation results for the FIFO algorithm with 4 frames clearly demonstrate the conditions that lead to **Belady's Anomaly**.

With a frame size of 4, the system experienced 10 page faults and achieved a low hit ratio of 0.17. Logically, one would expect that increasing physical memory (frames) would decrease the number of page faults. However, if this same reference string were run with only 3 frames, it would result in only 9 page faults.

This anomaly occurs because FIFO blindly replaces the oldest page in memory without considering how frequently or recently that page will be used. In this 4-frame scenario, pages 1 and 2 are kept in memory longer than they would be in a 3-frame setup. This causes the algorithm to evict pages that are actually needed sooner (like 3, 4, and 5), leading to a cascade of unnecessary faults. This proves that FIFO's lack of awareness regarding page usage patterns makes it highly inefficient and unpredictable for certain reference strings.

Conclusion :

FIFO is the simplest page replacement algorithm to implement, as it only requires a basic queue to track the order of page arrivals. However, this lab proves that simplicity comes at a severe cost. The demonstration of **Belady's Anomaly** shows that increasing physical memory can paradoxically degrade performance by increasing page faults. Because FIFO ignores the frequency and recency of page usage, it is generally outperformed by more intelligent algorithms like LRU (Least Recently Used) or Optimal Page Replacement in real-world operating systems, which are specifically designed to avoid this exact anomaly.

Github repository : https://github.com/fhmahi957/CSE_402_Assignment_materials.git